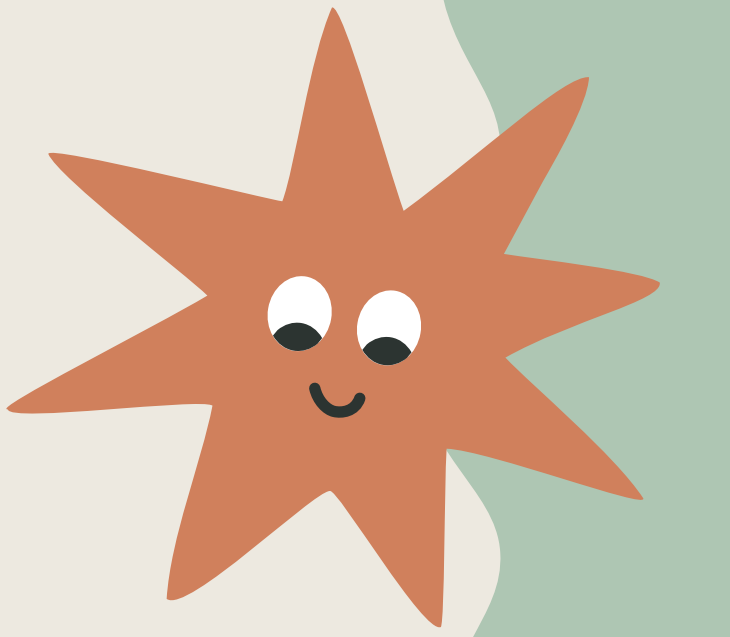
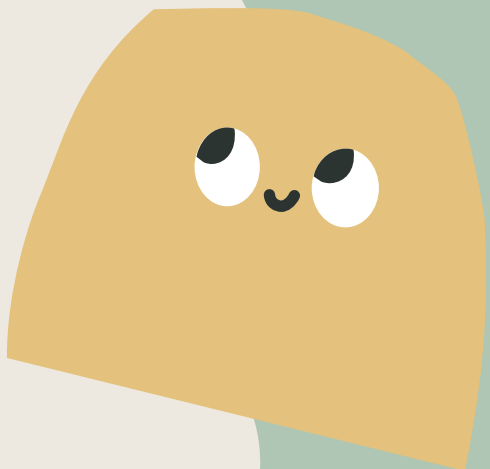
The Role of Swear Words in Toxicity Detection

How does the presence or absence of swear words affect the performance of toxicity detection systems?





LT2326 Machine learning for statistical NLP: Advanced
Eugene Wong
2026

Agenda

1. The Problem
2. Dataset & Lexicon
3. Evaluation Framework
4. Naïve Baselines
5. Classical ML vs Transformer
6. Future Work
7. Conclusion

Two Comments, Two Moderation Failures

“This article is fucking amazing!”

Excitement

but

Flagged as Toxic ❌

False alarm

“You better watch your back when you walk home tonight.”

Threat

but

Passed as Non-toxic ❌

Blind spot

Dataset & Lexicon

Toxicity Corpus

Kaggle Jigsaw Toxic Comment Classification

Human-labeled Wikipedia comments

Multi-label → Binary ground truth
(*toxic*, *severe_toxic*, *obscene*, *threat*, *insult*, *identity_hate*) → *is_toxic*

Removed unlabeled test data
labels value of -1

Profanity Lexicon

Sourced from noswearing.com

Whitelisted identity & anatomy
("gay", "lesbian", "queer", "anus", "penis", "testicle", "vagina", "mothafuckin\\\")

Cleanup

Token Masking
URLs and IPs

Remove Wikipedia markup
e.g. == *Section Heading* ==

Collapse elongated words
e.g. sooooo → soo

Data Overview

Dataset Key Metrics

159,571 train data

153,164 → **63,978** test data after filtering

Severe Class Imbalance

Train data

10.17% toxic comments (16,225)

Test data

9.76% toxic comments (6,243)

The 2x2 Sliced Evaluation Matrix (Test Data)

	Without Swears 57,870 **(90.45%)**	With Swears 6,108 **(9.55%)**
Non-toxic 57,735 **(90.24%)**	**Non-toxic without swears** 54,861 **(85.75%)**	**Non-toxic with swears** 2,874 **(4.49%)**
Toxic 6,243 **(9.76%)**	**Toxic without swears** 3,009 **(4.70%)**	**Toxic with swears** 3,234 **(5.05%)**

51.80% of toxic comments contain swear words
(3,234 out of 6,243)

4.98% of non-toxic comments contain swear words
(2,874 out of 57,735)

Two Metrics

ΔFPR

FPR(Non-toxic with Swears)

minus

FPR(Non-toxic without Swears)

Measures: False alarms

Intuition:
How much does casual swearing inflate false alarms?

Target: Ideal ΔFPR = 0.0%
Casual swearing should never trigger a false alarm.

ΔRecall

Recall(Toxic with Swears)

minus

Recall(Toxic without Swears)

Measures: Blind Spot

Intuition:
How much does detection drop when there are no swear words?

Target: Ideal ΔRecall = 0.0%
Identify toxic comments without swear words.

Naive Baselines

Majority Baseline

Always predicts "Non-Toxic"

Results

Accuracy: **90.24%**

Recall: **0%**

F1: **0%**

Rule-Based Lexicon Filter

Flag as “Toxic” when comment contains swear words

Results

Accuracy: **90.80%**

Recall(Toxic with Swears): **100%**
Recall(Toxic without Swears): **0%**
Unable to identify toxic comments when there are no swear words

FPR (Non-toxic with Swears): **100%**
FPR (Non-toxic without Swears): **0%**
Unable to recognize non-toxic comments when there are swear words

Results Comparison

TF-IDF + LR VS DeBERTa

F1:	57.93% VS 66.18% ✅
Recall (toxic without swear):	84.65% VS 88.60% ✅
FPR (non-toxic with swear):	65.00% VS 57.72% ✅
ΔRecall:	10.72% VS 8.21% ✅
ΔFPR:	54.61% VS 50.76% ✅

Surprisingly, both perform similarly

Even a transformer struggles to recognize non-toxic comments when they contain swear words (ΔFPR is high).

Why Both Models Struggle with Swears

TF-IDF + LR

Feature weights * LR coefficients

fuck **+22.72**
fucking **+19.76**
stupid **+18.67**
shit **+18.03**
idiot **+15.87**

Swear words have too much weight. Any swear word instantly triggers toxicity

DeBERTa

Train data imbalance

78.22% of comments with swear words are toxic

This leads to the model taking the easy way out.

Examples

False Alarm

"Well honestly I cant be fucking bothered tonight. So sleep wee all u editors ans admits . I'm off to bed. CU anaon"

Human label: **Not toxic**

Has swear words: **Yes**

TF-IDF + LR prediction: **Toxic**
99.43% chance of being toxic

DeBERTa prediction: **Toxic**
99.61% chance of being toxic

Blind Spot

"Nice job inserting your left-wing garbage in Jan Brewers article. I removed it and if you vandalize the article again you will be blocked from editing on Wikipedia."

Human label: **Toxic**

Has swear words: **No**

TF-IDF + LR prediction: **Non-toxic**
47.32 % chance of being toxic

DeBERTa prediction: **Non-toxic**
8.63% chance of being toxic

IMPROVEMENTS

Data Augmentation

Adding swear words to non-toxic comments without changing the intent and toxicity

"This is great!" → "This is fucking great!"

More non-toxic with swear words samples in the training data.

Adjusting Weights in the Loss

Adjust loss weights to account for data class imbalance

Bigger penalty for missing toxic comments without swear words.

Bigger penalty for falsely flagging non-toxic comments with swear words as toxic

7. CONCLUSION

Takeaways

Swear words ≠ Toxicity

PRESENCE of swear words (false alarm)
DeBERTa ΔFPR: **50.76%**
False alarms on innocent comments surge by **50.76%** when swear words are present

ABSENCE of swear words (blind spot)
DeBERTa ΔRecall: **8.21%**
Detection of toxic comments drops by **8.21%** when no swear words are present

"Effective moderation requires understanding conversational intent, not reacting to vocabulary."

